

# QUBO Formulation: Weighted Max-2-SAT

## Weighted Max-2-SAT

**Problem Statement.** Given a set of  $N$  Boolean variables  $\mathcal{X} = \{x_1, \dots, x_N\}$  and a collection of  $M$  weighted clauses  $\mathcal{C} = \{C_1, \dots, C_M\}$ , where each clause  $C_j$  contains at most two literals and carries a non-negative weight  $w_j \geq 0$ , the Weighted Max-2-SAT problem seeks a truth assignment to  $\mathcal{X}$  that maximizes the total weight of satisfied clauses.

**Decision Variables.** For each Boolean variable  $x_i \in \{0, 1\}$ ,  $i \in \{1, \dots, N\}$ , we introduce a binary decision variable  $x_i \in \{0, 1\}$  representing its truth value (1 for true, 0 for false). Each clause  $C_j$  involves a set of literals  $\mathcal{L}_j$ . For each literal  $\ell \in \mathcal{L}_j$ , we define a binary indicator  $y_{j,\ell} \in \{0, 1\}$  that evaluates to 1 if the literal is true under the assignment and 0 otherwise. Specifically, if  $\ell = x_i$ , then  $y_{j,\ell} = x_i$ ; if  $\ell = \neg x_i$ , then  $y_{j,\ell} = 1 - x_i$ .

**QUBO Derivation. Step 1.** The original objective is to maximize the total weight of satisfied clauses:

$$\max_{\mathbf{x} \in \{0,1\}^N} \sum_{j=1}^M w_j s_j, \quad (1)$$

where  $s_j \in \{0, 1\}$  is the satisfaction indicator for clause  $C_j$ . For a clause with two literals  $\ell_{j,1}$  and  $\ell_{j,2}$ , the indicator is given by the logical OR:

$$s_j = 1 - (1 - y_{j,\ell_{j,1}})(1 - y_{j,\ell_{j,2}}) = y_{j,\ell_{j,1}} + y_{j,\ell_{j,2}} - y_{j,\ell_{j,1}}y_{j,\ell_{j,2}}. \quad (2)$$

Since QUBO solvers minimize energy, we define the Hamiltonian as the negative of the objective:

$$H(\mathbf{x}) = - \sum_{j=1}^M w_j s_j = - \sum_{j=1}^M w_j (y_{j,\ell_{j,1}} + y_{j,\ell_{j,2}} - y_{j,\ell_{j,1}}y_{j,\ell_{j,2}}). \quad (3)$$

**Step 2.** The problem is an unconstrained maximization over the binary hypercube  $\{0, 1\}^N$ . There are no explicit constraints on the variable assignments.

**Step 3.** Consequently, no penalty terms are required. The feasible space is the entire domain  $\{0, 1\}^N$ , and the Hamiltonian directly encodes the objective function.

**Step 4.** Combining the objective terms yields the single QUBO Hamiltonian:

$$H(\mathbf{x}) = \sum_{j=1}^M w_j (-y_{j,\ell_{j,1}} - y_{j,\ell_{j,2}} + y_{j,\ell_{j,1}} y_{j,\ell_{j,2}}). \quad (4)$$

Substituting the literal mappings  $y_{j,\ell} \in \{x_i, 1 - x_i\}$  and expanding using the idempotent property  $x_i^2 = x_i$ , each clause  $C_j$  contributes linear, quadratic, and constant terms to  $H(\mathbf{x})$ . For a clause involving variables  $x_i$  and  $x_k$  with polarities  $p_{j,1}, p_{j,2} \in \{0, 1\}$  (where 0 denotes a positive literal and 1 denotes a negated literal), the expansion yields:

$$H_j(\mathbf{x}) = -w_j(1 - 2p_{j,1})x_i - w_j(1 - 2p_{j,2})x_k + w_j(1 - 2p_{j,1})(1 - 2p_{j,2})x_i x_k - w_j(1 - p_{j,1} - p_{j,2} + p_{j,1}p_{j,2}). \quad (5)$$

Summing over all clauses gives the full Hamiltonian  $H(\mathbf{x}) = \sum_{j=1}^M H_j(\mathbf{x})$ .

**Step 5.** Reading off the coefficients from the expanded Hamiltonian, the QUBO matrix entries  $Q_{i,k}$  and offset  $c$  are given in closed form by:

$$Q_{i,i} = \sum_{j: x_i \in C_j} -w_j(1 - 2p_{j,m}), \quad (6)$$

$$Q_{i,k} = \sum_{j: x_i, x_k \in C_j} w_j(1 - 2p_{j,1})(1 - 2p_{j,2}), \quad (7)$$

$$c = \sum_{j=1}^M -w_j(1 - p_{j,1} - p_{j,2} + p_{j,1}p_{j,2}), \quad (8)$$

where  $p_{j,m}$  is the polarity of the  $m$ -th literal in clause  $C_j$  mapping to variable  $x_i$  or  $x_k$ . For clauses with only one literal, the corresponding quadratic and second linear terms vanish.

**Penalty Weight Justification.** The Weighted Max-2-SAT problem is inherently unconstrained over the binary assignment space. Therefore, no penalty weights  $\lambda$  are introduced, and the penalty term set is empty. The Hamiltonian  $H(\mathbf{x})$  directly represents the negative of the objective function, eliminating the need for constraint penalties or weight tuning.

**Correctness Argument.** Minimizing  $H(\mathbf{x})$  over  $\{0, 1\}^N$  is mathematically equivalent to maximizing  $\sum_{j=1}^M w_j s_j$ . Since  $w_j \geq 0$  and  $s_j \in \{0, 1\}$ , each term  $-w_j s_j$  in the Hamiltonian is minimized when  $s_j = 1$ . Any assignment that leaves a clause  $C_j$  unsatisfied ( $s_j = 0$ ) results in a strictly higher energy contribution (0) compared to a satisfied assignment ( $-w_j$ ). Consequently, the global minimum of  $H(\mathbf{x})$  must occur at a truth assignment that maximizes the total satisfied clause weight, guaranteeing that the QUBO ground state corresponds to the optimal solution of the original problem.

**Illustrative Example.** Consider a concrete instance with  $N = 3$  variables and  $M = 2$  clauses. Let  $C_1 = (x_1 \vee x_2)$  with weight  $w_1 = 5$ , and  $C_2 = (\neg x_2 \vee x_3)$  with weight  $w_2 = 3$ .

For  $C_1$ , both literals are positive ( $p_{1,1} = 0, p_{1,2} = 0$ ). Applying the general formula:

$$H_1(\mathbf{x}) = -5x_1 - 5x_2 + 5x_1x_2. \quad (9)$$

For  $C_2$ , the first literal is negated ( $p_{2,1} = 1$ ) and the second is positive ( $p_{2,2} = 0$ ). Applying the general formula:

$$H_2(\mathbf{x}) = -3(1-2)x_2 - 3(1)x_3 + 3(1-2)(1)x_2x_3 - 3(1-1-0+0) = 3x_2 - 3x_3 - 3x_2x_3. \quad (10)$$

Summing  $H_1$  and  $H_2$  gives the concrete Hamiltonian:

$$H(\mathbf{x}) = -5x_1 - 2x_2 - 3x_3 + 5x_1x_2 - 3x_2x_3. \quad (11)$$

The corresponding QUBO matrix entries and offset are:

$$Q_{1,1} = -5, \quad Q_{2,2} = -2, \quad Q_{3,3} = -3, \quad (12)$$

$$Q_{1,2} = 5, \quad Q_{2,3} = -3, \quad Q_{1,3} = 0, \quad (13)$$

$$c = 0. \quad (14)$$

Evaluating  $H(\mathbf{x})$  over the  $2^3 = 8$  possible assignments reveals that the assignment  $\mathbf{x}^* = (1, 0, 1)$  yields the minimum energy  $H(\mathbf{x}^*) = -8$ . This corresponds to satisfying both clauses, achieving the maximum possible weight of  $5 + 3 = 8$ .